Multiple Testing and Simulation-Based Inference

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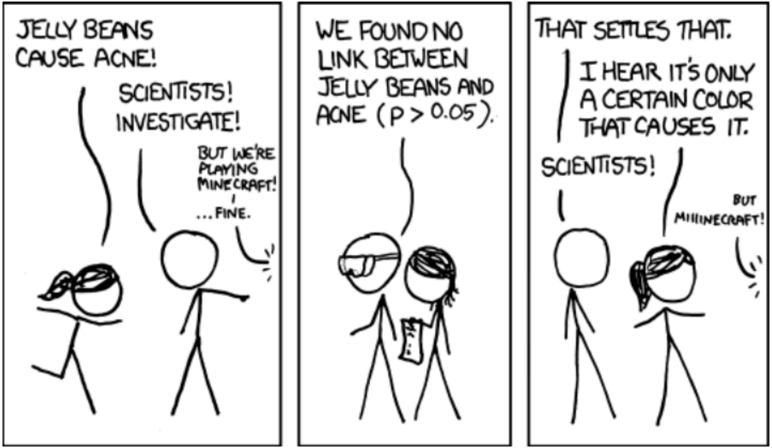
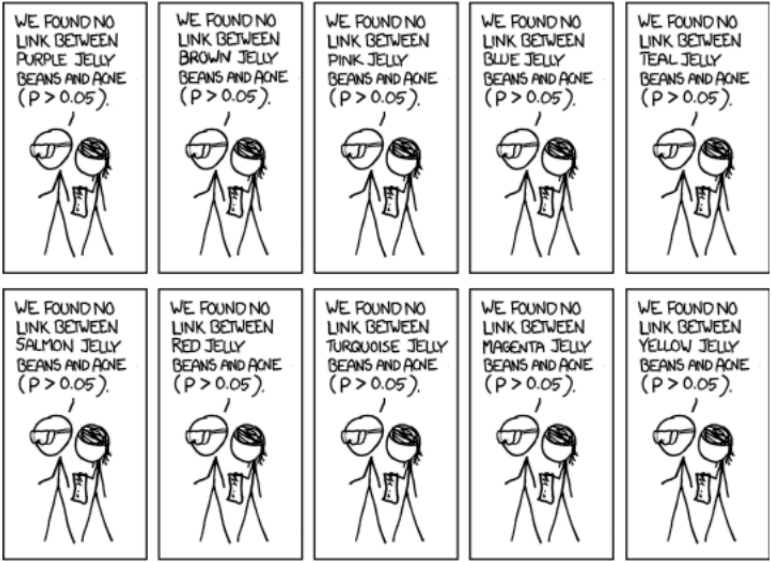
Introduction

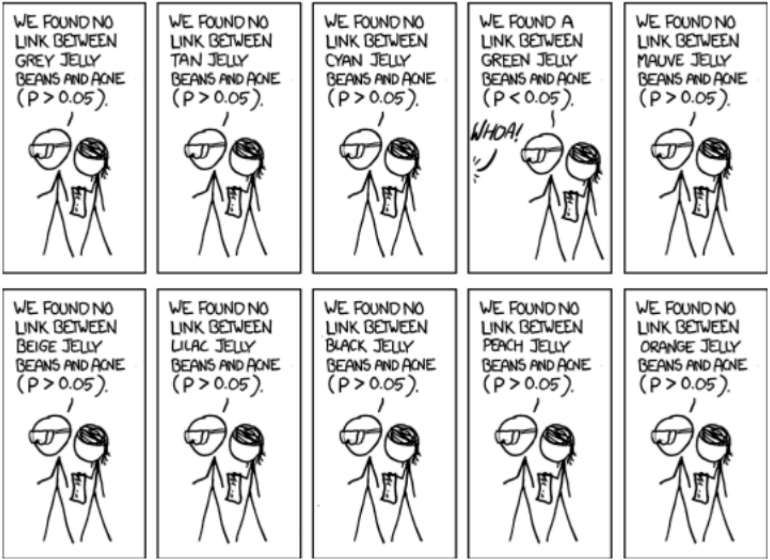
Do jelly beans cause acne?



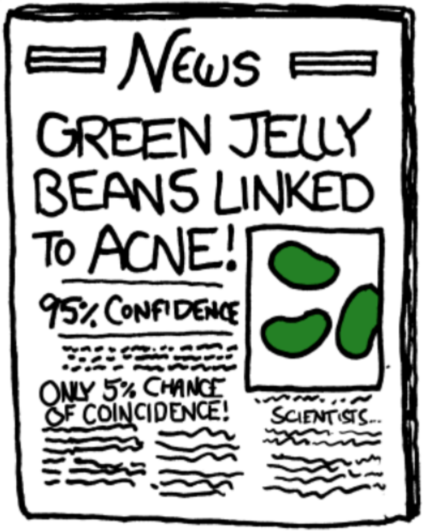
Twenty colors, twenty tests



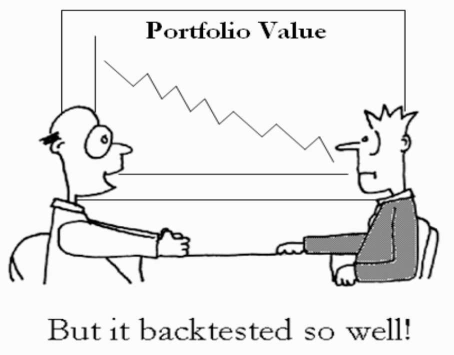
One of them twinkles



The headline writes itself



Closer to home



Suhonen, Lennkh & Perez (2017)

J. Portfolio Management

215 commercially promoted alternative-beta strategies.

In backtest:
median Sharpe ratio $\approx$ **1.2**

Out-of-sample:
median Sharpe ratio $\approx$ **0.3**

Median deterioration: 73%.

Backtested SR overstates out-of-sample SR by 4$\times$

*This is **data snooping**: the jelly-bean mechanism applied to quantitative finance at industrial scale.*
The reported backtest SR is the maximum over a large family of implicit tests.

A typical regression table

Body fat data

Garcia et al. (2005); $n = 71$ healthy German women

Example from Hothorn, Bretz & Westfall (2008).

- Response: `DEXfat` (body fat from Dual-Energy X-Ray Absorptiometry, the expensive “gold standard”)
- $p = 9$ cheap anthropometric predictors: waist, hip, elbow, knee, and four log-skinfold composites

Question. Can we predict DXA body fat from cheap measurements? `lmod <- lm(DEXfat ~ ., data = bodyfat)`

`summary(lmod)`

	Est.	*t stat*	*p-value*
age	0.02	0.62	0.538
waistcirc	0.21	3.14	0.003**
hipcirc	0.34	4.27	<0.001***
elbowbreadth	−0.41	−0.40	0.688
kneebreadth	1.76	2.43	0.018*
anthro3a	5.74	1.10	0.274
anthro3b	9.87	1.74	0.086·
anthro3c	0.39	0.19	0.853
anthro4	−6.57	−1.01	0.315

$R^2 = 0.92$, $F_{9,61} = 81.4$, $p < 10^{-29}$

· $p < 0.10$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

Let’s do what everyone does: **search for stars**.

The innocent question

Four flagged variables.

`waistcirc`** `hipcirc`*** `kneebreadth`* (`anthro3b`.)

The global F-test is overwhelming:

$$F_{9,61} = 81.4, \qquad p < 10^{-29}$$

Something in the model is real.

How many of these stars should we believe?

- What error rate does "scan the table, flag the stars" actually control?
- Is it the 5% we think we're operating at?

Stress-testing the rule

Strip the signal. Same regression machinery, no truth to find.

$$y_i = \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0,1), \qquad X \in \mathbb{R}^{n \times k} \text{ independent of } y.$$

All slopes $\beta_1, \ldots, \beta_k$ are *exactly* zero by construction.

Ask two questions at the 5% level, on every simulated dataset:

1. Does *any* of the k slope $|t_j|$ exceed the critical value? (`reject_any`)
2. Does the joint F-test of $\beta_1 = \cdots = \beta_k = 0$ reject? (`reject_F`)

Under correct size, *both* should reject $\approx 5\%$ of the time.

Demo 0: The cost of searching for stars

`demo_intro.R`

- Simulate $M = 10{,}000$ datasets under the pure null: all slopes zero.
- Two rejection rules at 5%: any significant t vs. joint F-test.
- Increase k and watch `reject_any` climb while `reject_F` holds at 0.05.
- The gap between the two rates is the cost of searching.

The F-test controls size because it makes one decision, not k.
Searching for stars makes k decisions and pays no multiplicity price.

The simulation reveal

```r
set.seed(1)
M <- 20000; n <- 50; k <- 10; alpha <- 0.05
reject_any <- reject_F <- logical(M)

for (m in seq_len(M)) {
  X <- matrix(rnorm(n * k), n, k)
  y <- rnorm(n)                          # null: y independent of X
  fit <- summary(lm(y ~ X))
  t_stats <- fit$coefficients[-1, "t value"]
  crit    <- qt(1 - alpha/2, df = n - k - 1)
  reject_any[m] <- any(abs(t_stats) > crit)
  f <- fit$fstatistic
  reject_F[m]   <- pf(f[1], f[2], f[3], lower.tail = FALSE) < alpha
}
c(any_t = mean(reject_any), F = mean(reject_F))
```

The simulation reveal

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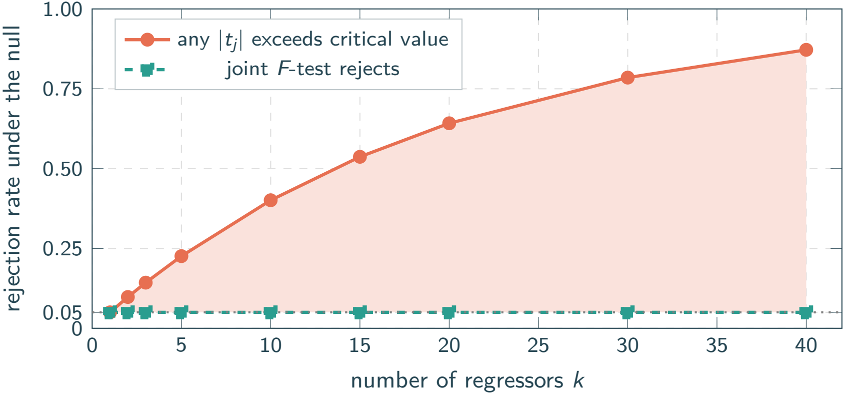
}
c(any_t = mean(reject_any), F = mean(reject_F))
```

$$\texttt{any\_t} = \mathbf{0.373} \qquad \texttt{F} = \mathbf{0.052}$$

This is what searching for stars costs.

How bad does it get?

Sweep k. Fixed $n = 100$, nominal $\alpha = 0.05$.



The shaded region is the **cost of searching**. The joint F-test tests all slopes simultaneously under a single null and controls size at α regardless of k, but it cannot identify which β_j is nonzero.

Searching for stars, jointly

Same data, same estimates, same t-statistics. Different reference distribution.

	Est.	t	unadj. p	adj. p
age	0.02	0.62	0.538	0.996
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hipcirc	0.34	4.27	<0.001***	0.001**
elbowbreadth	−0.41	−0.40	0.688	1.000
kneebreadth	1.76	2.43	0.018*	0.131
anthro3a	5.74	1.10	0.274	0.895
anthro3b	9.87	1.74	0.086.	0.478
anthro3c	0.39	0.19	0.853	1.000
anthro4	−6.57	−1.01	0.315	0.930

Of the three starred variables, **`kneebreadth` flips**. `waistcirc` and `hipcirc` survive, though with larger adjusted p-values.

Adjusted p-values use the joint distribution of the nine t-statistics (single-step max-t). *More on this later.*

Hothorn, Bretz & Westfall (2008), *Biometrical Journal* 50(3), §6.2.

“Why not just look at the first table?”

Per row, in isolation. Every unadjusted p-value is a valid 5%-level statement about *that specific* coefficient, if chosen in advance.

Searching. Reading the whole table and flagging the stars is *one* procedure applied to all rows at once. Under the null,

$$P(\text{at least one star}) \approx 1-(1-\alpha)^k = 40\% \text{ for } k=10.$$

A procedure that advertises 5% and delivers **40%**.

Noticing. Reporting only the row you “happened to notice” doesn’t escape the problem. You walked up to `kneebreadth` *because* its p-value was small; the other eight rows were the competition. The unadjusted $p=0.018$ answers a question you didn’t ask.

Unadjusted is valid *per row*. Adjusted is valid *for the table*.

Two days, four parts

Part I

Foundations

- The testing family; error rates (PCER, FWER, FDR)
- Closure
- Bonferroni, Holm
- Geometry of multiple tests

Demo: body fat regression

Part II

Simulation-based methods

- Max-t in the linear model
- Monte Carlo tests
- When pivotality fails
- Joint vs. marginal inference

Demo: MC max-t; US GDP nonlinearity

Part III

Beyond FWER

- k-FWER, FDP, FDR
- MC combination tests
- Cauchy combination test
- Applied illustration: Clements (2024)

Demo: sparse vs. dense alternatives

Part IV

Model Confidence Set

- Model selection under uncertainty
- The superior set $\mathcal{M}^*$
- Bootstrap implementation
- Horse race: US CPI inflation

Demo: US CPI, 10 models, MCS package

Each session is accompanied by a live R demo.

We came to search for stars.
We will leave knowing how to find them.